

NOMBRE

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MATERIA

INF-021-50 TELECOMUNICACIONES
[SEPT-DIC 2026] (PUNTA CANA)

PROYECTO

RUTAS ESTÁTICAS EN CADENA (4 VMS)

PROFESOR

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MANDATO

Tarea: Implementar en **GNS3** una red de tres sedes conectadas en cadena por tres **routers**, con direccionamiento **IPv4** y rutas estáticas, usando **4 máquinas virtuales Windows** como terminales. El trabajo pide demostrar la conectividad extrema a extremo, filtrar tráfico con una **ACL** extendida y diagnosticar una falla provocada en la tabla de enrutamiento.

Entregar en la fecha establecida en un documento **PDF**.

INTRODUCION

Para esta práctica levanté en **GNS**3 una red de tres sedes conectadas en cadena: tres **routers Cisco**, tres **switches** y cuatro **máquinas virtuales Windows 10** como terminales. Todo el tráfico entre sedes se mueve con rutas estáticas, escritas a mano en cada router. Sobre esa base hay dos puntos que le dan peso al laboratorio: una **ACL** extendida en el router central que bloquea el ping de una **LAN** hacia otra, y una falla provocada en la tabla de enrutamiento para ver cómo se comporta la red cuando le falta una ruta y cómo se diagnostica eso.

DESARROLLO

La topología usa tres routers (**R1, R2 y R3**) unidos en cadena por enlaces punto a punto **/30**, y tres switches, uno por sede. De **SW1** cuelgan **PC-A1** y **PC-A2**, de **SW2** la **PC-B1** y de **SW3** la **PC-C1**. Todo el tráfico entre sedes pasa sí o sí por **R2**.

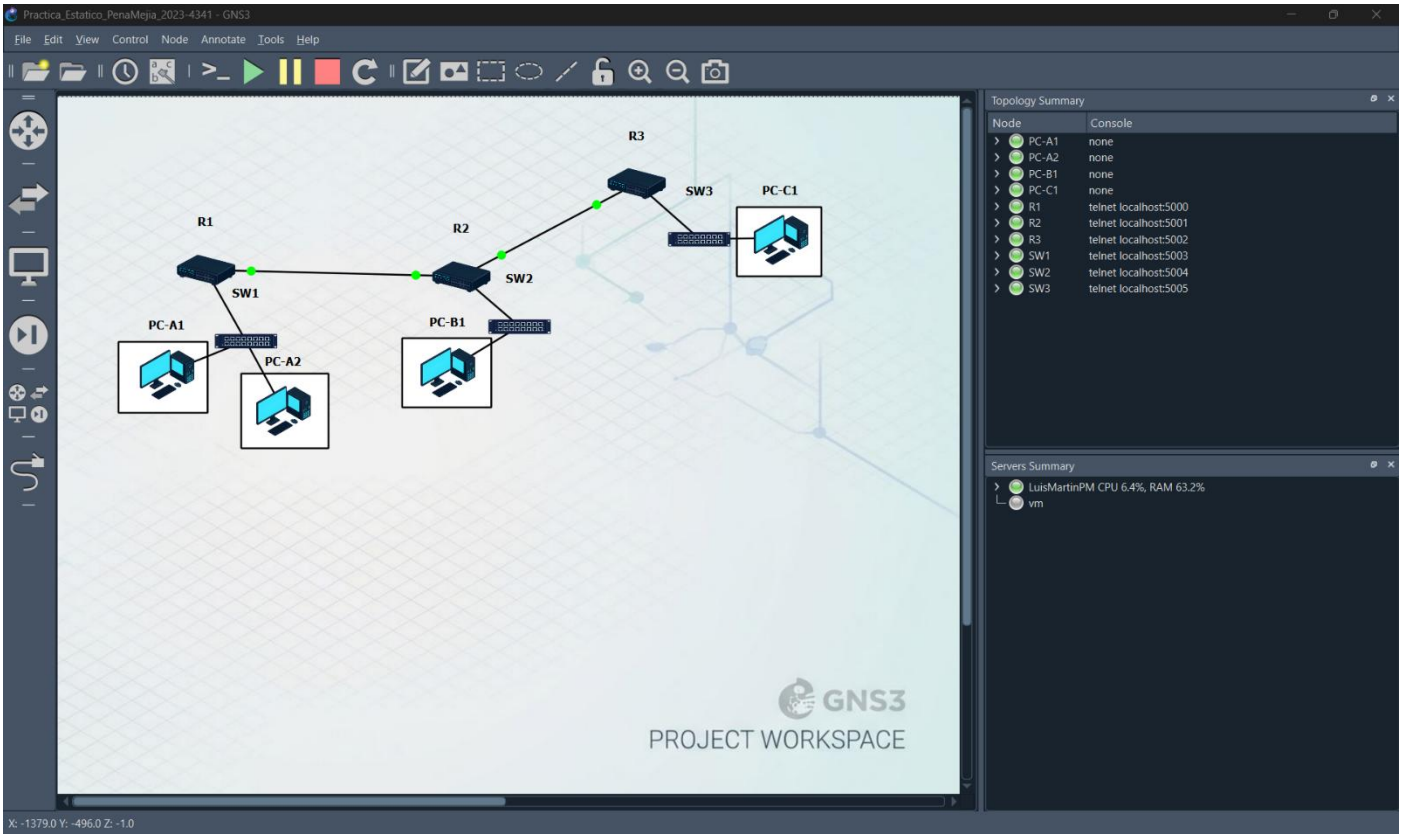


TABLA 1. PLAN DE DIRECCIONAMIENTO DE LA RED					
RED	DISPOSITIVO	INTERFAZ	IP	MÁSCARA	GATEWAY
LAN-A	R1	Fa0/0	192.168.10.1	255.255.255.0	–
LAN-A	PC-A1	Ethernet	192.168.10.11	255.255.255.0	192.168.10.1
LAN-A	PC-A2	Ethernet	192.168.10.12	255.255.255.0	192.168.10.1
LAN-B	R2	Fa0/0	192.168.20.1	255.255.255.0	–
LAN-B	PC-B1	Ethernet	192.168.20.11	255.255.255.0	192.168.20.1
LAN-C	R3	Fa0/0	192.168.30.1	255.255.255.0	–
LAN-C	PC-C1	Ethernet	192.168.30.11	255.255.255.0	192.168.30.1
WAN-1	R1	Fa0/1	10.0.12.1	255.255.255.252	–
WAN-1	R2	Fa0/1	10.0.12.2	255.255.255.252	–
WAN-2	R2	Fa1/0	10.0.23.1	255.255.255.252	–
WAN-2	R3	Fa0/1	10.0.23.2	255.255.255.252	–
Se usaron interfaces FastEthernet en lugar de GigabitEthernet porque esa es la nomenclatura real de la imagen Cisco 3725 usada en la emulación.					

VERIFICACIÓN DE PUERTOS EN LOS SWITCHES

SW1

```
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#no ip domain-lookup
SW1(config)#line console 0
SW1(config-line)#logging synchronous
SW1(config-line)#end
SW1#
*Mar  1 00:11:07.815: %SYS-5-CONFIG_I: Configured from console by console
SW1#write memory
Building configuration...
[OK]
SW1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            unassigned      YES unset  administratively down down

FastEthernet0/1            unassigned      YES unset  administratively down down

FastEthernet1/0            unassigned      YES unset  up                    up

FastEthernet1/1            unassigned      YES unset  up                    up

FastEthernet1/2            unassigned      YES unset  up                    up

FastEthernet1/3            unassigned      YES unset  up                    down

FastEthernet1/4            unassigned      YES unset  up                    down

FastEthernet1/5            unassigned      YES unset  up                    down

FastEthernet1/6            unassigned      YES unset  up                    down

FastEthernet1/7            unassigned      YES unset  up                    down

FastEthernet1/8            unassigned      YES unset  up                    down

 --More--
```

SW2

```
et1/6, changed state to down
SW2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#no ip domain-lookup
SW2(config)#line console 0
SW2(config-line)#logging synchronous
SW2(config-line)#end
SW2#
*Mar  1 00:10:55.759: %SYS-5-CONFIG_I: Configured from console by console
SW2#write memory
Building configuration...
[OK]
SW2#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            unassigned      YES unset  administratively down down

FastEthernet0/1            unassigned      YES unset  administratively down down

FastEthernet1/0            unassigned      YES unset  up                    up

FastEthernet1/1            unassigned      YES unset  up                    up

FastEthernet1/2            unassigned      YES unset  up                    down

FastEthernet1/3            unassigned      YES unset  up                    down

FastEthernet1/4            unassigned      YES unset  up                    down

FastEthernet1/5            unassigned      YES unset  up                    down

FastEthernet1/6            unassigned      YES unset  up                    down

FastEthernet1/7            unassigned      YES unset  up                    down

FastEthernet1/8            unassigned      YES unset  up                    down

 --More--
```

SW3

```
et1/6, changed state to down
SW3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#no ip domain-lookup
SW3(config)#line console 0
SW3(config-line)#logging synchronous
SW3(config-line)#end
SW3#
*Mar  1 00:12:21.791: %SYS-5-CONFIG_I: Configured from console by console
SW3#write memory
Building configuration...
[OK]
SW3#show ip interface brief
Interface                  IP-Address      OK? Method Status                Prot
ocol
FastEthernet0/0            unassigned      YES unset  administratively down down

FastEthernet0/1            unassigned      YES unset  administratively down down

FastEthernet1/0            unassigned      YES unset  up                    up

FastEthernet1/1            unassigned      YES unset  up                    up

FastEthernet1/2            unassigned      YES unset  up                    down

FastEthernet1/3            unassigned      YES unset  up                    down

FastEthernet1/4            unassigned      YES unset  up                    down

FastEthernet1/5            unassigned      YES unset  up                    down

FastEthernet1/6            unassigned      YES unset  up                    down

FastEthernet1/7            unassigned      YES unset  up                    down

FastEthernet1/8            unassigned      YES unset  up                    down

 --More--
```

CONFIGURACIÓN DE LOS DISPOSITIVOS

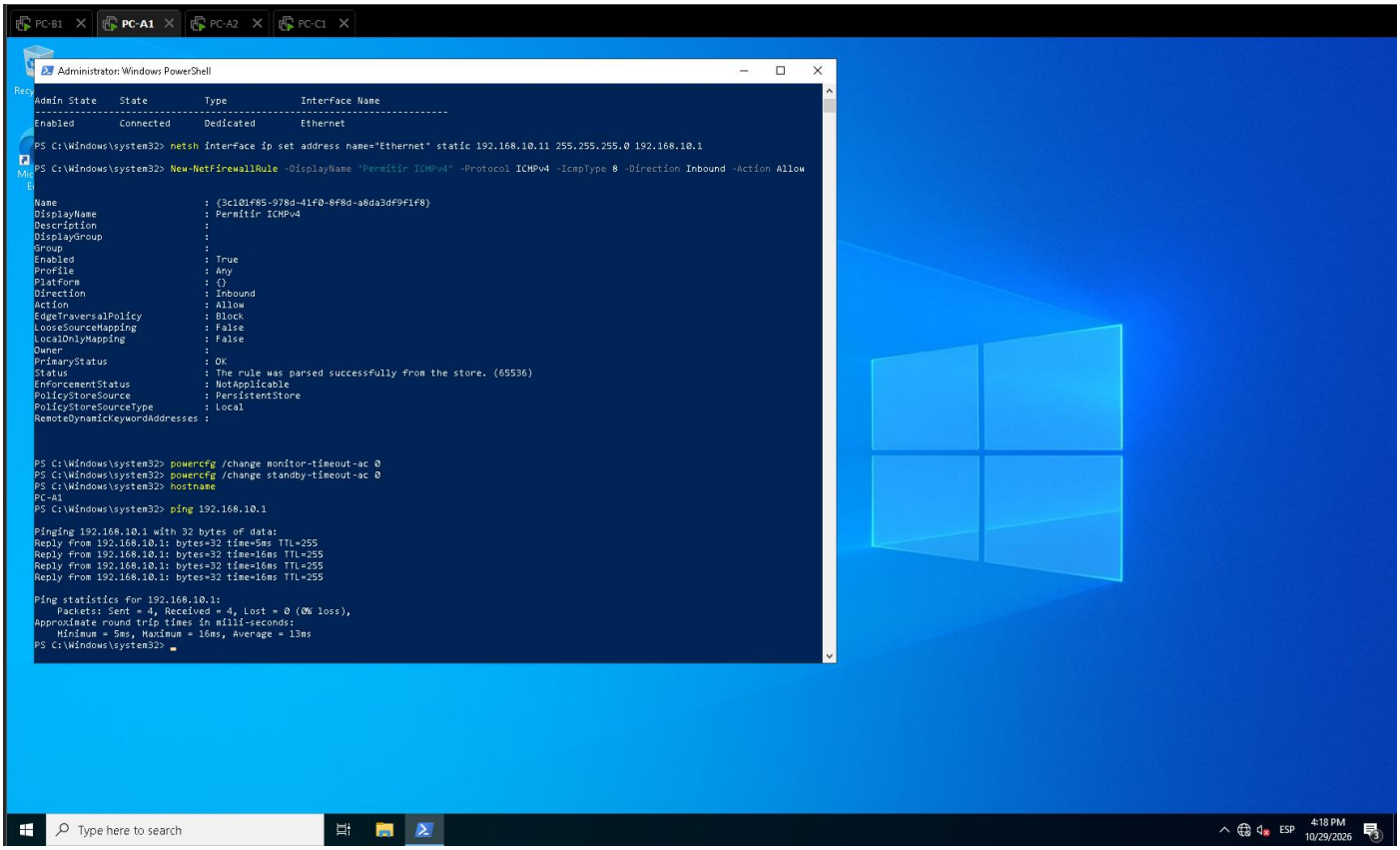
Las configuraciones finales de cada equipo salieron del comando ***show running-config***. Profe la subí aquí en **GitHub**, perdón si no respeté lo que pidió, pero es que en mi diseñador interior se ve horrible pegar los textos así crudos y tener **13 páginas** para un **TXT** es ilógico, por lo menos para mi así que lo puse más lindo aquí, le dije a la **"Primo"** que me lo ponga lindo.

https://github.com/luismartin023/TELECOMUNICACIONES

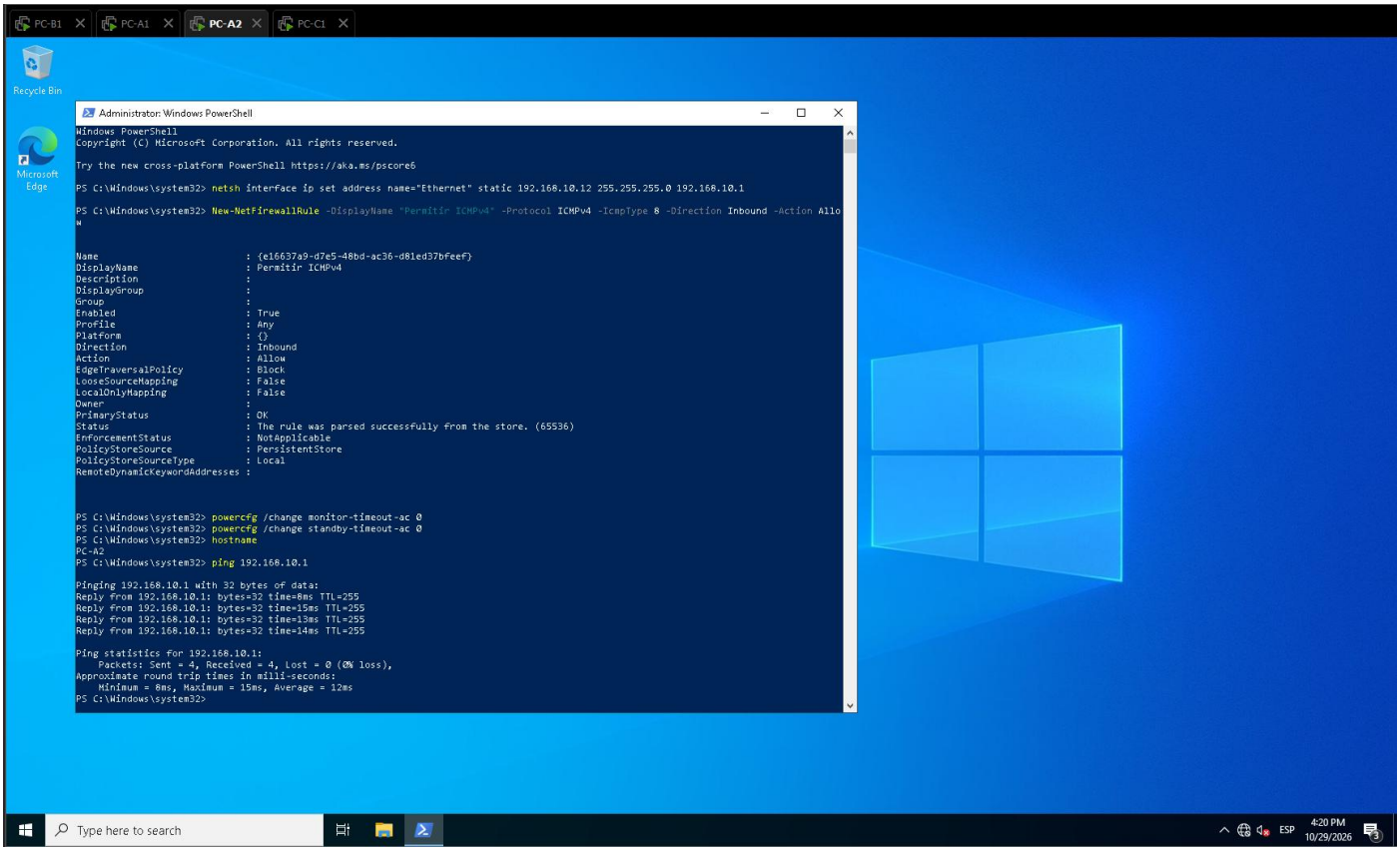
CONFIGURACIÓN DE LAS VMs

Cada **VM** recibió su propio hostname, una **IP** estática según su **LAN** y una regla en el firewall para aceptar **ICMPv4** entrante, que es lo que permite que el ping responda. Las cuatro salieron del mismo Windows base como clones enlazados: se instaló una sola vez y **GNS3** generó las copias.

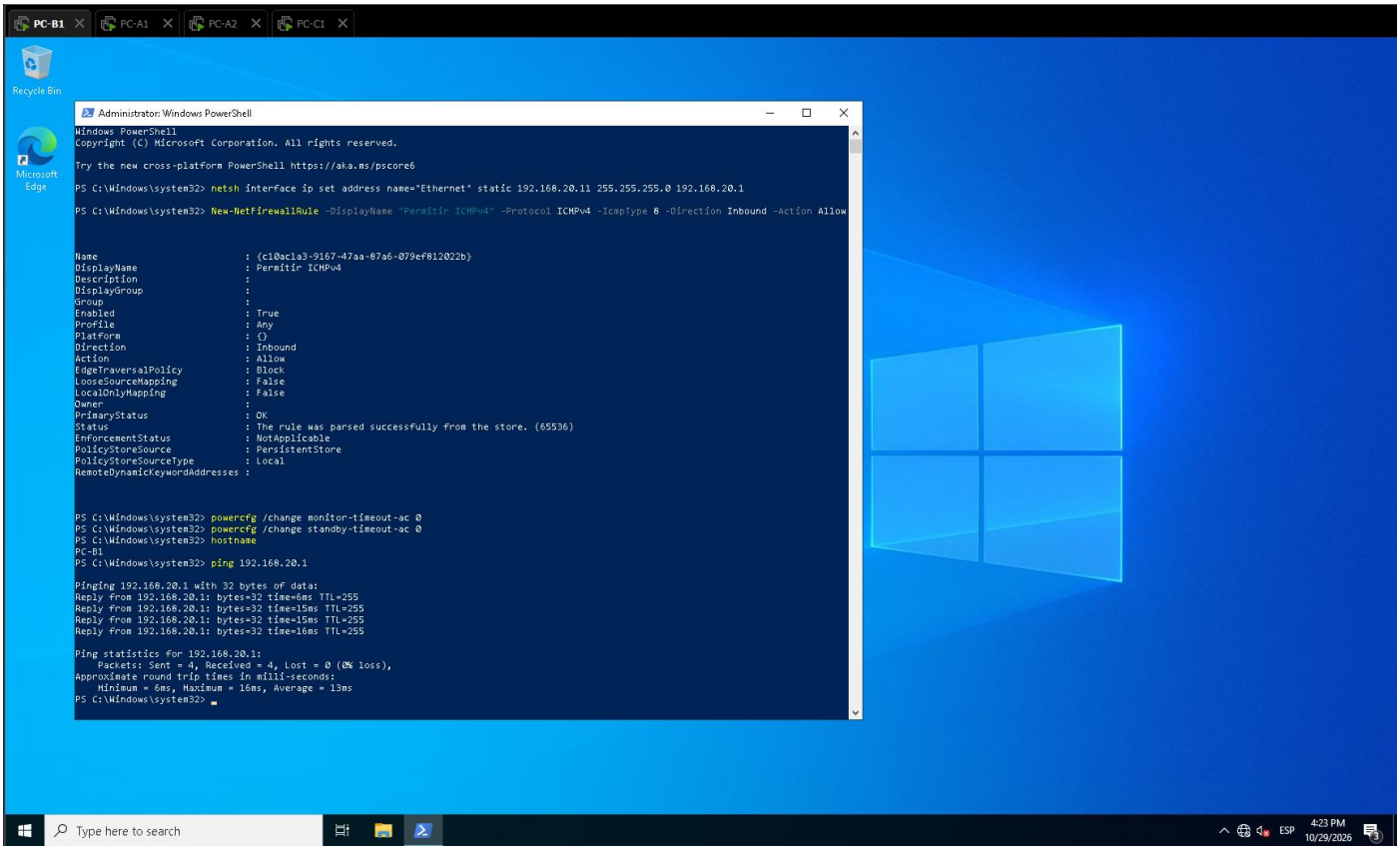
PC-A1



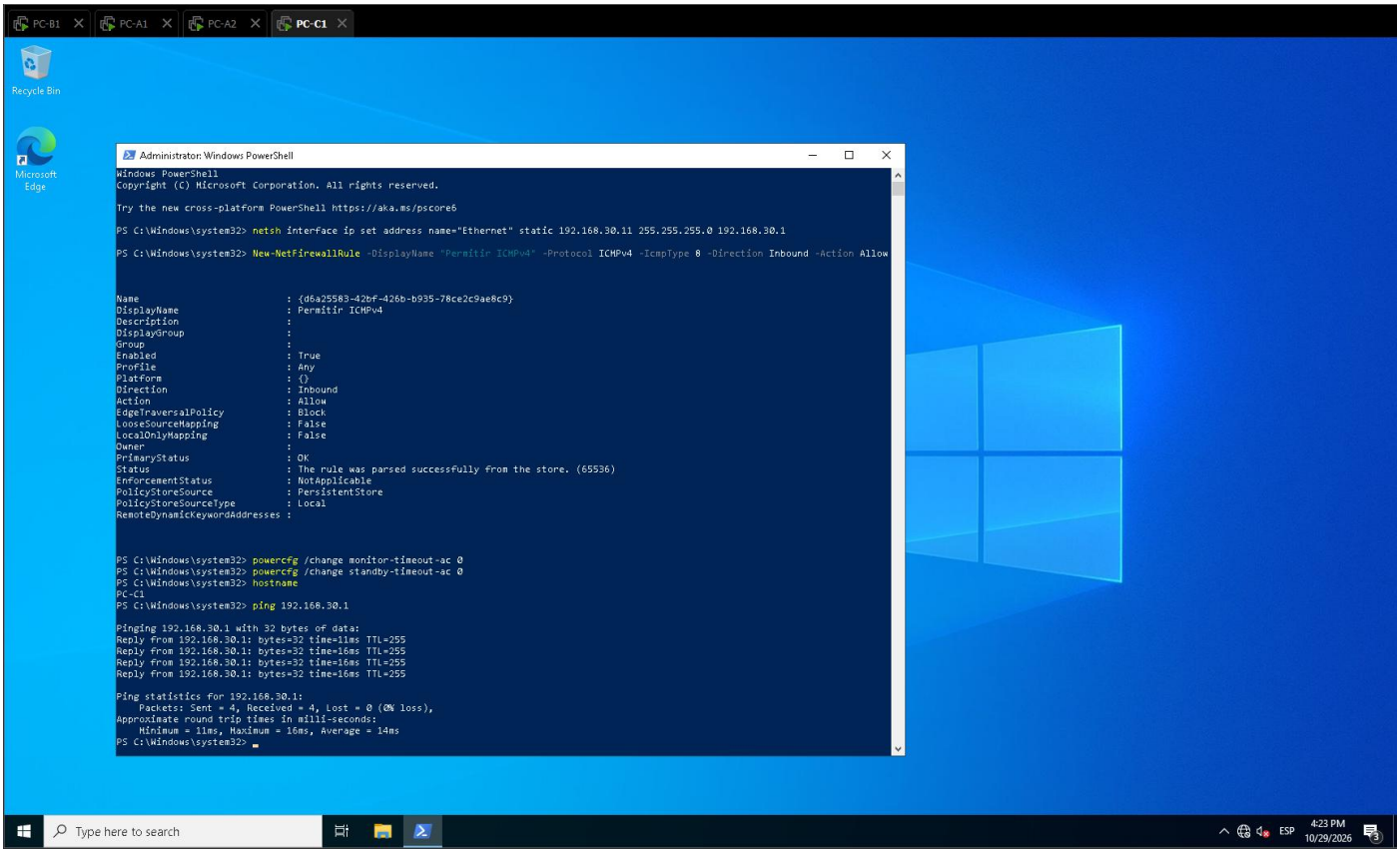
PC-A2


PC-B1
PC-A1
PC-A2
PC-C1
Recycle Bin
Microsoft Edge
Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> netsh interface ip set address name="Ethernet" static 192.168.10.12 255.255.255.0 192.168.10.1
PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Permitir ICMPv4" -Protocol ICMPv4 -IcmpType 8 -Direction Inbound -Action Allow
Name : {e16637a9-d7e5-48bd-ac36-d81ed37bfeef}
DisplayName : Permitir ICMPv4
Description :
DisplayGroup :
Group :
Enabled : True
Profile : Any
Platform : {}
Direction : Inbound
Action : Allow
EdgeTraversalPolicy : Block
LooseSourceMapping : False
LocalOnlyMapping : False
Owner :
PrimaryStatus : OK
Status : The rule was parsed successfully from the store. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PS C:\Windows\system32> powercfg /change monitor-timeout-ac 0
PS C:\Windows\system32> powercfg /change standby-timeout-ac 0
PS C:\Windows\system32> hostname
PC-A2
PS C:\Windows\system32> ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=8ms TTL=255
Reply from 192.168.10.1: bytes=32 time=15ms TTL=255
Reply from 192.168.10.1: bytes=32 time=13ms TTL=255
Reply from 192.168.10.1: bytes=32 time=14ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 15ms, Average = 12ms
PS C:\Windows\system32>
Type here to search
ESP
4:20 PM
10/29/2026

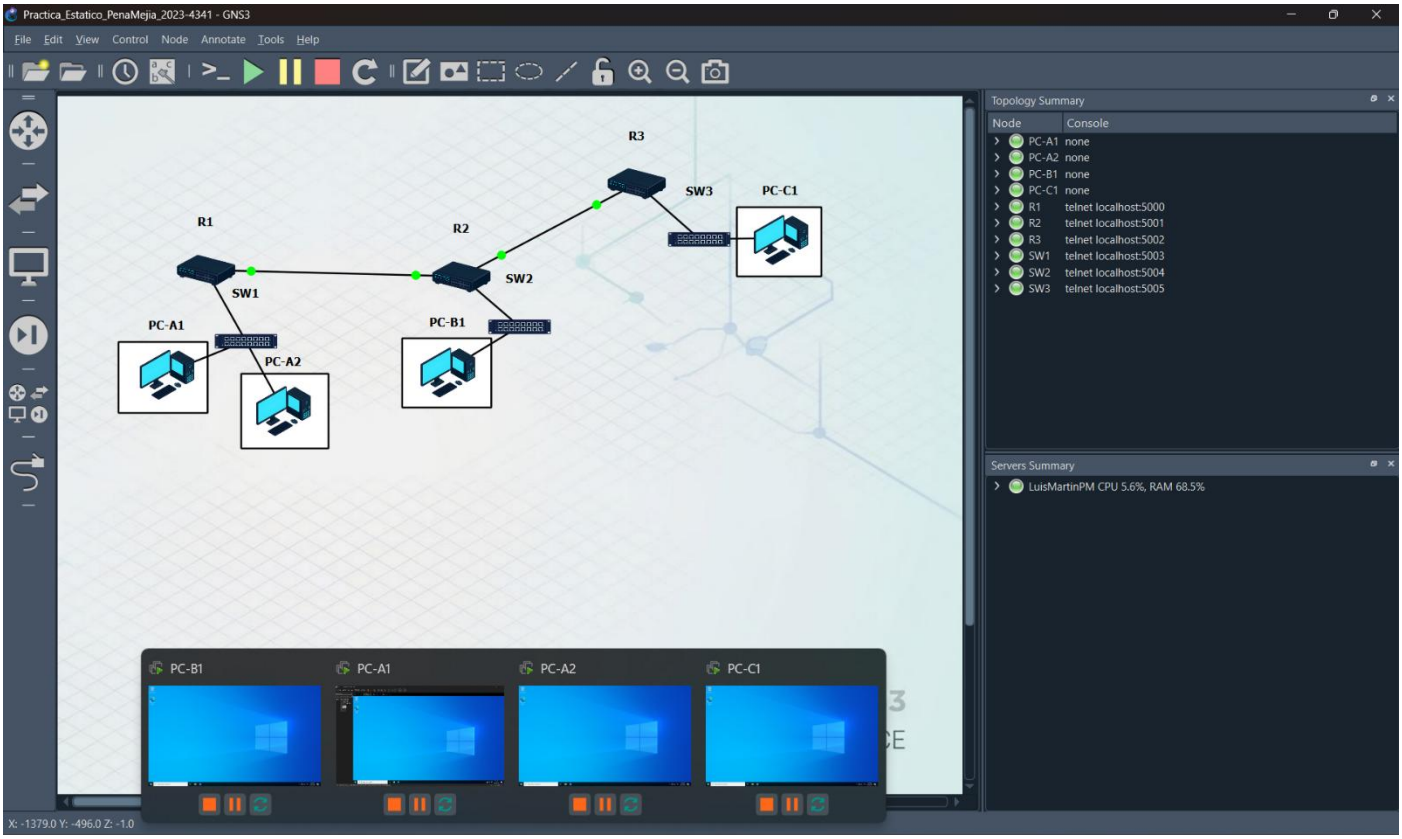
PC-B1


PC-B1
PC-A1
PC-A2
PC-C1
Recycle Bin
Microsoft Edge
Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> netsh interface ip set address name="Ethernet" static 192.168.20.11 255.255.255.0 192.168.20.1
PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Permitir ICMPv4" -Protocol ICMPv4 -IcmpType 8 -Direction Inbound -Action Allow
Name : {c10ac1a3-9167-47aa-87a6-079ef812022b}
DisplayName : Permitir ICMPv4
Description :
DisplayGroup :
Group :
Enabled : True
Profile : Any
Platform : {}
Direction : Inbound
Action : Allow
EdgeTraversalPolicy : Block
LooseSourceMapping : False
LocalOnlyMapping : False
Owner :
PrimaryStatus : OK
Status : The rule was parsed successfully from the store. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PS C:\Windows\system32> powercfg /change monitor-timeout-ac 0
PS C:\Windows\system32> powercfg /change standby-timeout-ac 0
PS C:\Windows\system32> hostname
PC-B1
PS C:\Windows\system32> ping 192.168.20.1
Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time=6ms TTL=255
Reply from 192.168.20.1: bytes=32 time=15ms TTL=255
Reply from 192.168.20.1: bytes=32 time=15ms TTL=255
Reply from 192.168.20.1: bytes=32 time=16ms TTL=255
Ping statistics for 192.168.20.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 16ms, Average = 13ms
PS C:\Windows\system32>
Type here to search
ESP
4:23 PM
10/29/2026

PC-C1


PC-B1
PC-A1
PC-A2
PC-C1
Recycle Bin
Microsoft Edge
Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Windows\system32> netsh interface ip set address name="Ethernet" static 192.168.30.11 255.255.255.0 192.168.30.1
PS C:\Windows\system32> New-NetFirewallRule -DisplayName "Permitir ICMPv4" -Protocol ICMPv4 -IcmpType 8 -Direction Inbound -Action Allow
Name : {d6a25583-42bf-426b-b935-78ce2c9ae8c9}
DisplayName : Permitir ICMPv4
Description :
DisplayGroup :
Group :
Enabled : True
Profile : Any
Platform : {}
Direction : Inbound
Action : Allow
EdgeTraversalPolicy : Block
LooseSourceMapping : False
LocalOnlyMapping : False
Owner :
PrimaryStatus : OK
Status : The rule was parsed successfully from the store. (65536)
EnforcementStatus : NotApplicable
PolicyStoreSource : PersistentStore
PolicyStoreSourceType : Local
RemoteDynamicKeywordAddresses :
PS C:\Windows\system32> powercfg /change monitor-timeout-ac 0
PS C:\Windows\system32> powercfg /change standby-timeout-ac 0
PS C:\Windows\system32> hostname
PC-C1
PS C:\Windows\system32> ping 192.168.30.1
Pinging 192.168.30.1 with 32 bytes of data:
Reply from 192.168.30.1: bytes=32 time=11ms TTL=255
Reply from 192.168.30.1: bytes=32 time=16ms TTL=255
Reply from 192.168.30.1: bytes=32 time=16ms TTL=255
Reply from 192.168.30.1: bytes=32 time=16ms TTL=255
Ping statistics for 192.168.30.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 11ms, Maximum = 16ms, Average = 14ms
PS C:\Windows\system32>
Type here to search
ESP
4:23 PM
10/29/2026

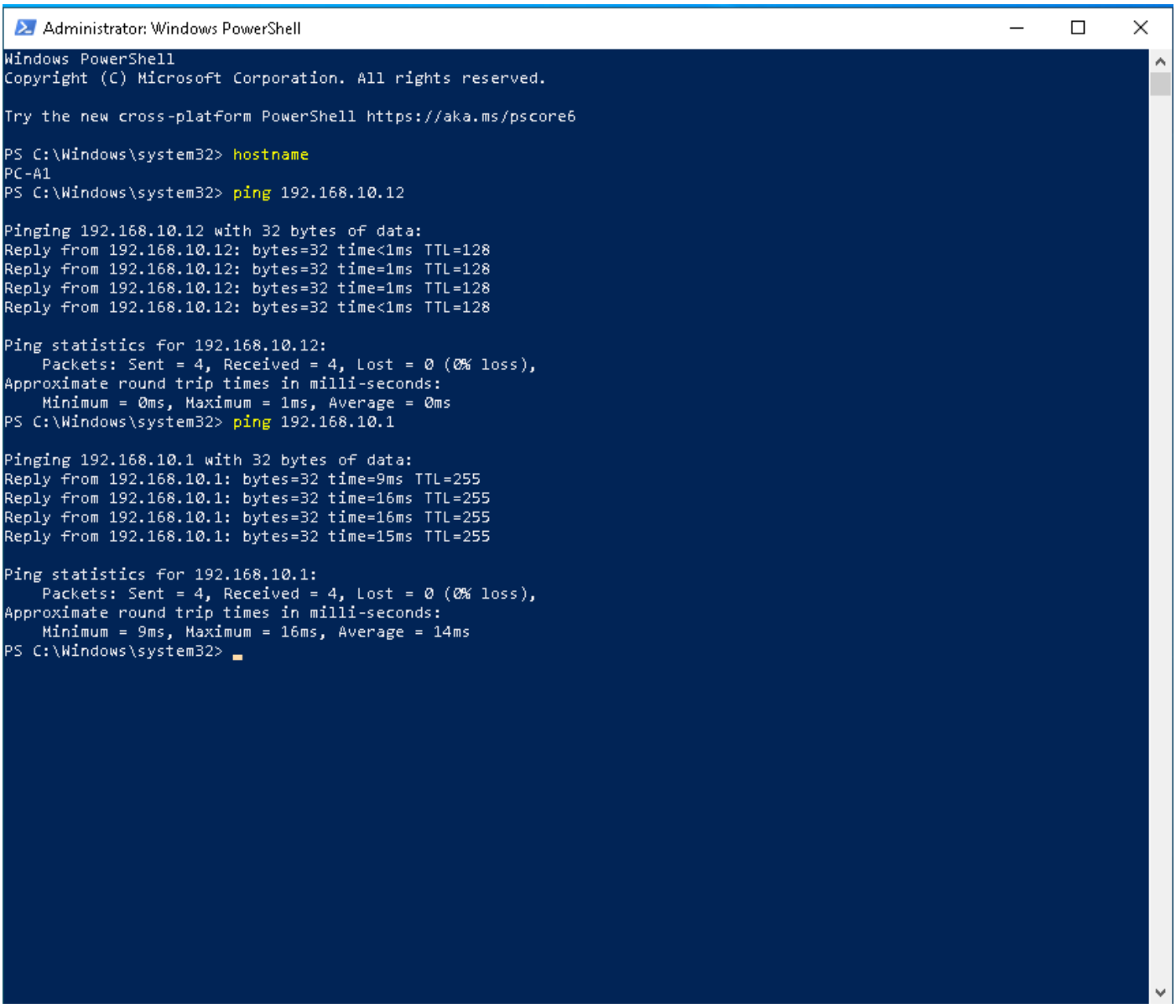
Las 4 VMs Windows corriendo en VMware



PRUEBAS

Conectividad local en LAN-A

ping de PC-A1 a PC-A2 y a su gateway, responde todo.



LAN-A hacia LAN-B

ping de PC-A1 a PC-B1 cruzando R1 y R2.

```
PS C:\Windows\system32> ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:
Reply from 192.168.20.11: bytes=32 time=54ms TTL=126
Reply from 192.168.20.11: bytes=32 time=62ms TTL=126
Reply from 192.168.20.11: bytes=32 time=64ms TTL=126
Reply from 192.168.20.11: bytes=32 time=64ms TTL=126

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 54ms, Maximum = 64ms, Average = 61ms
PS C:\Windows\system32>
```

Extremo a extremo

ping de PC-A1 a PC-C1 atravesando los 3 routers.

```
PS C:\Windows\system32> ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:
Reply from 192.168.30.11: bytes=32 time=86ms TTL=125
Reply from 192.168.30.11: bytes=32 time=94ms TTL=125
Reply from 192.168.30.11: bytes=32 time=78ms TTL=125
Reply from 192.168.30.11: bytes=32 time=93ms TTL=125

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 78ms, Maximum = 94ms, Average = 87ms
PS C:\Windows\system32>
```

Ruta seguida

tracert muestra los 4 saltos: 10.1 → 12.2 → 23.2 → destino

```
PS C:\Windows\system32> tracert 192.168.30.11

Tracing route to 192.168.30.11 over a maximum of 30 hops

  1    27 ms    14 ms    14 ms  192.168.10.1
  2    48 ms    45 ms    48 ms  10.0.12.2
  3    77 ms    77 ms    76 ms  10.0.23.2
  4    96 ms    76 ms    91 ms  192.168.30.11

Trace complete.
PS C:\Windows\system32>
```

Bloqueo por ACL

ping de PC-B1 a PC-C1 falla: la ACL de R2 lo corta.

```
PS C:\Windows\system32> hostname
PC-B1
PS C:\Windows\system32> ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:
Reply from 192.168.20.1: Destination net unreachable.
Reply from 192.168.20.1: Destination net unreachable.
Reply from 192.168.20.1: Destination net unreachable.
Reply from 192.168.20.1: Destination net unreachable.

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
PS C:\Windows\system32>
```

Sentido contrario libre

ping de PC-C1 a PC-B1 responde — la ACL solo filtra B→C.

```
PS C:\Windows\system32> hostname
PC-C1
PS C:\Windows\system32> ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:
Reply from 192.168.20.11: bytes=32 time=54ms TTL=126
Reply from 192.168.20.11: bytes=32 time=63ms TTL=126
Reply from 192.168.20.11: bytes=32 time=63ms TTL=126
Reply from 192.168.20.11: bytes=32 time=61ms TTL=126

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 54ms, Maximum = 63ms, Average = 60ms
PS C:\Windows\system32>
```

Tabla de rutas de R1

rutas conectadas (C) y estáticas (S).

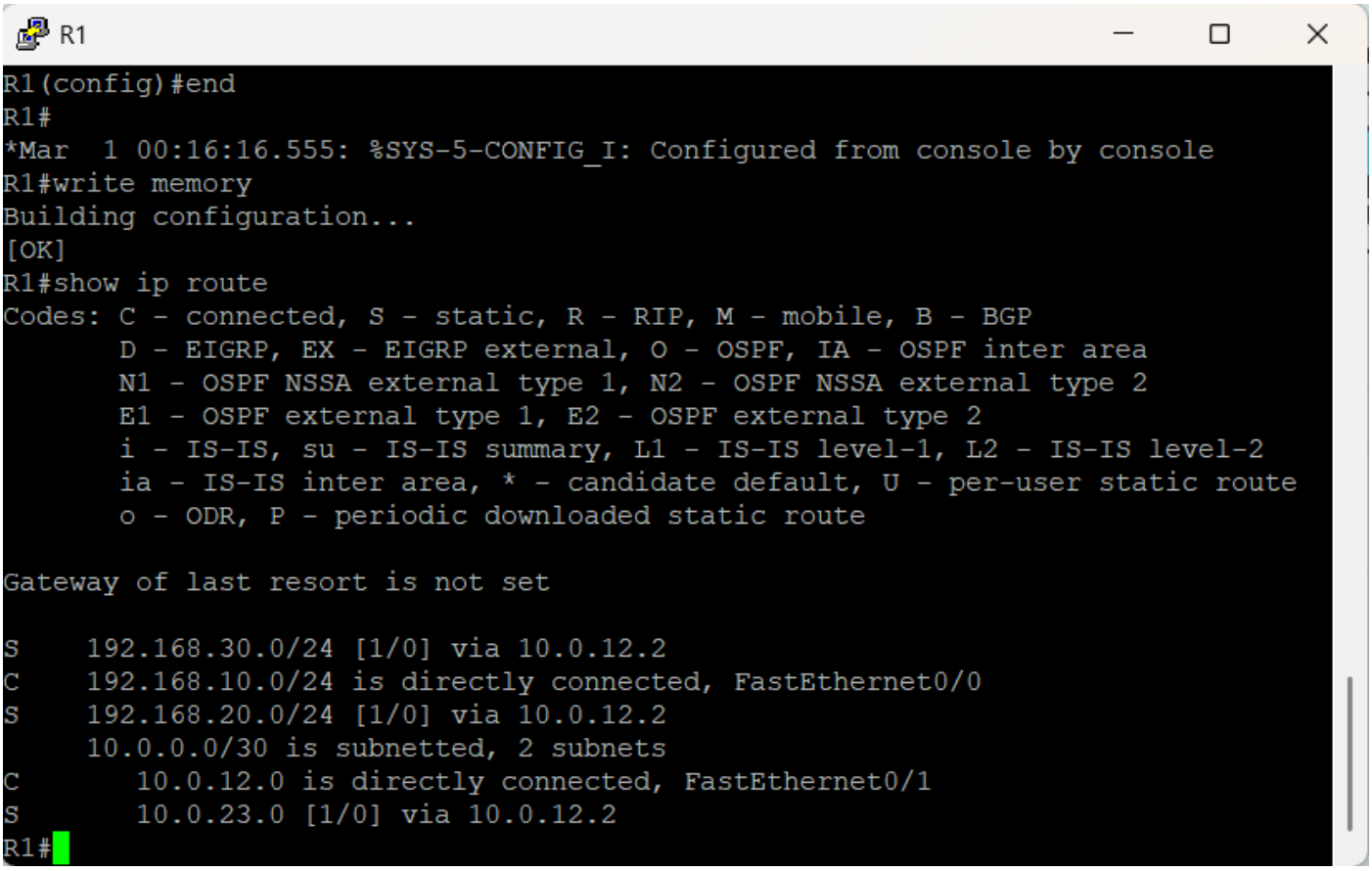
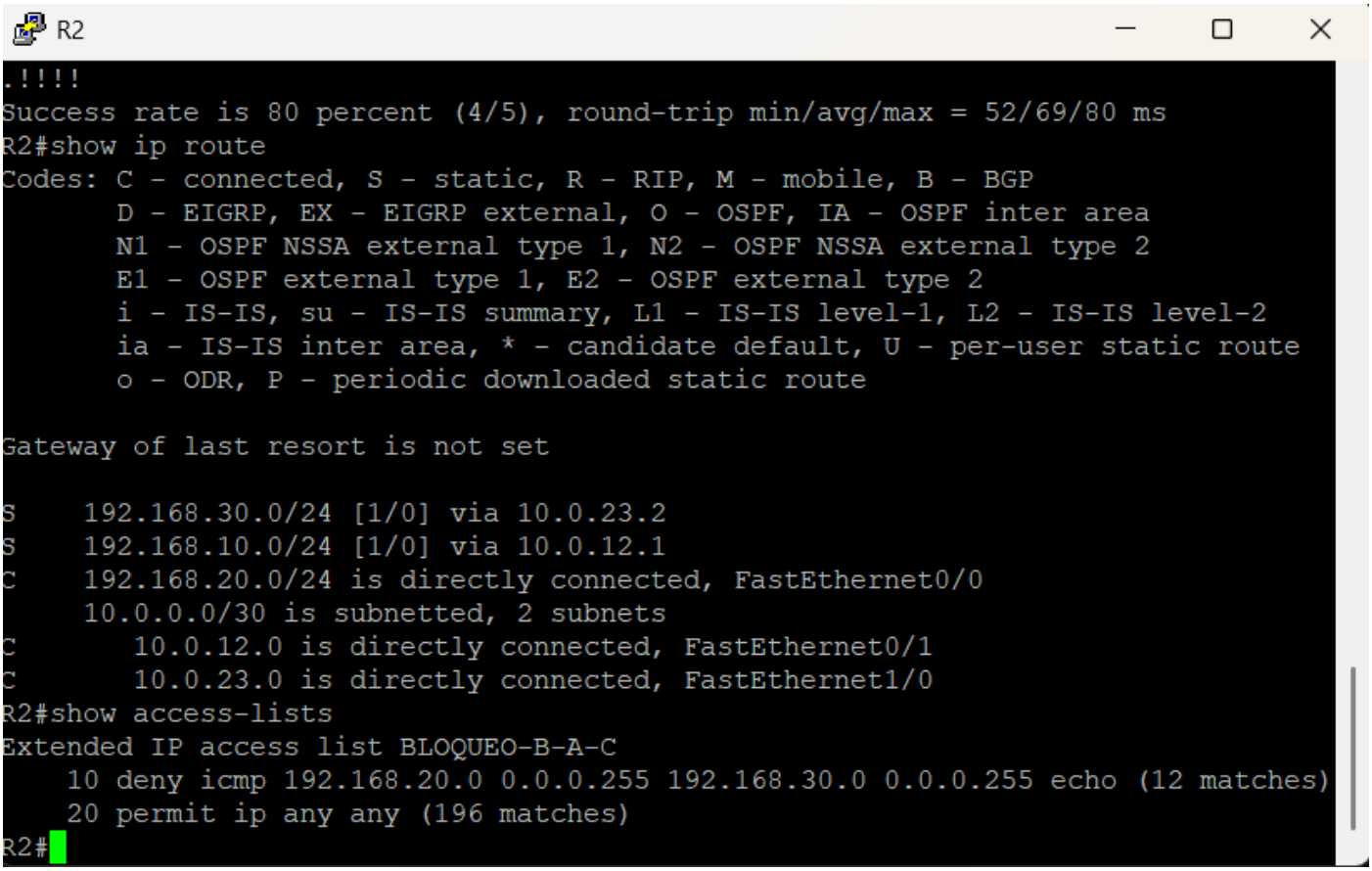


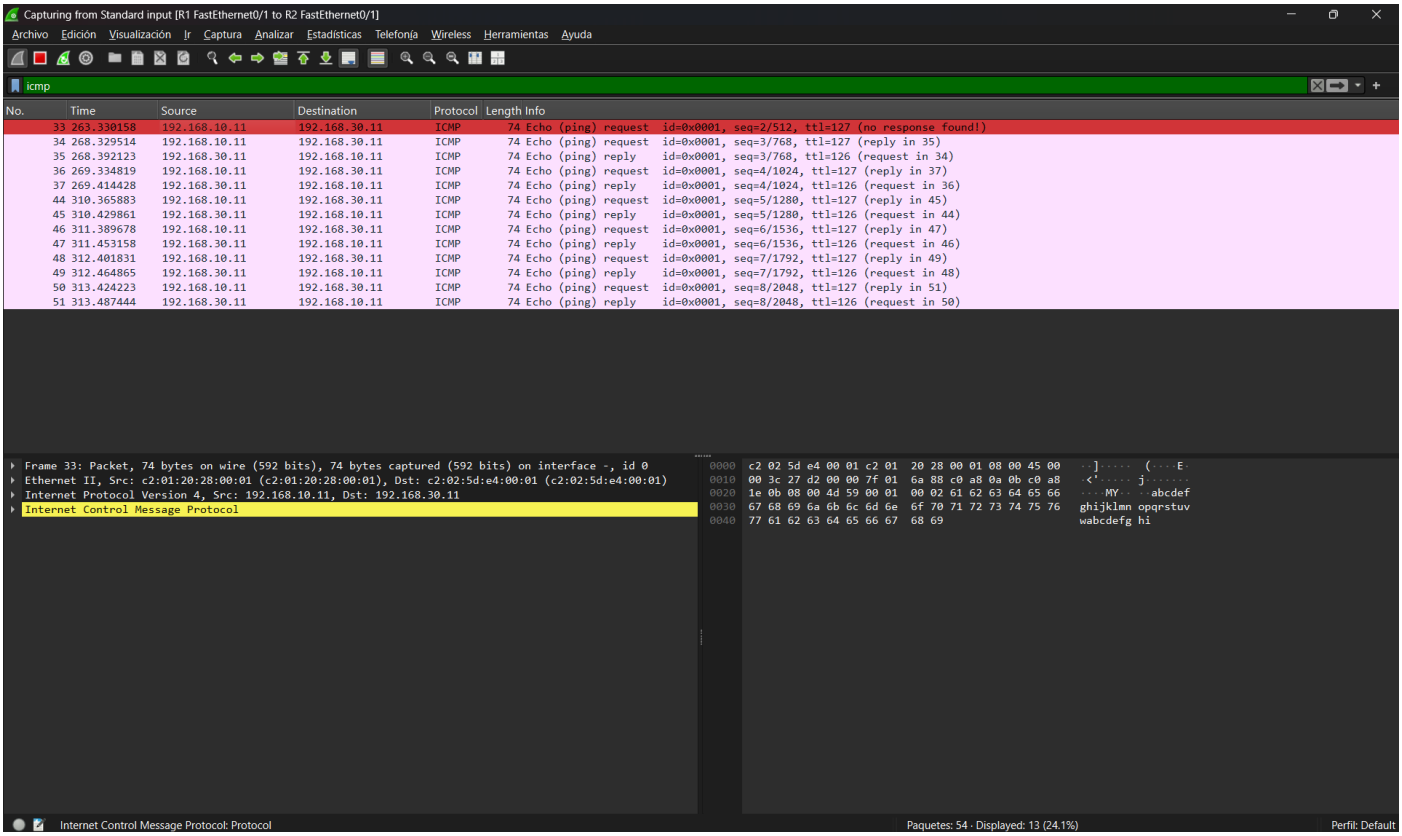
Tabla de rutas y ACL de R2

rutas correctas + contador del deny con coincidencias



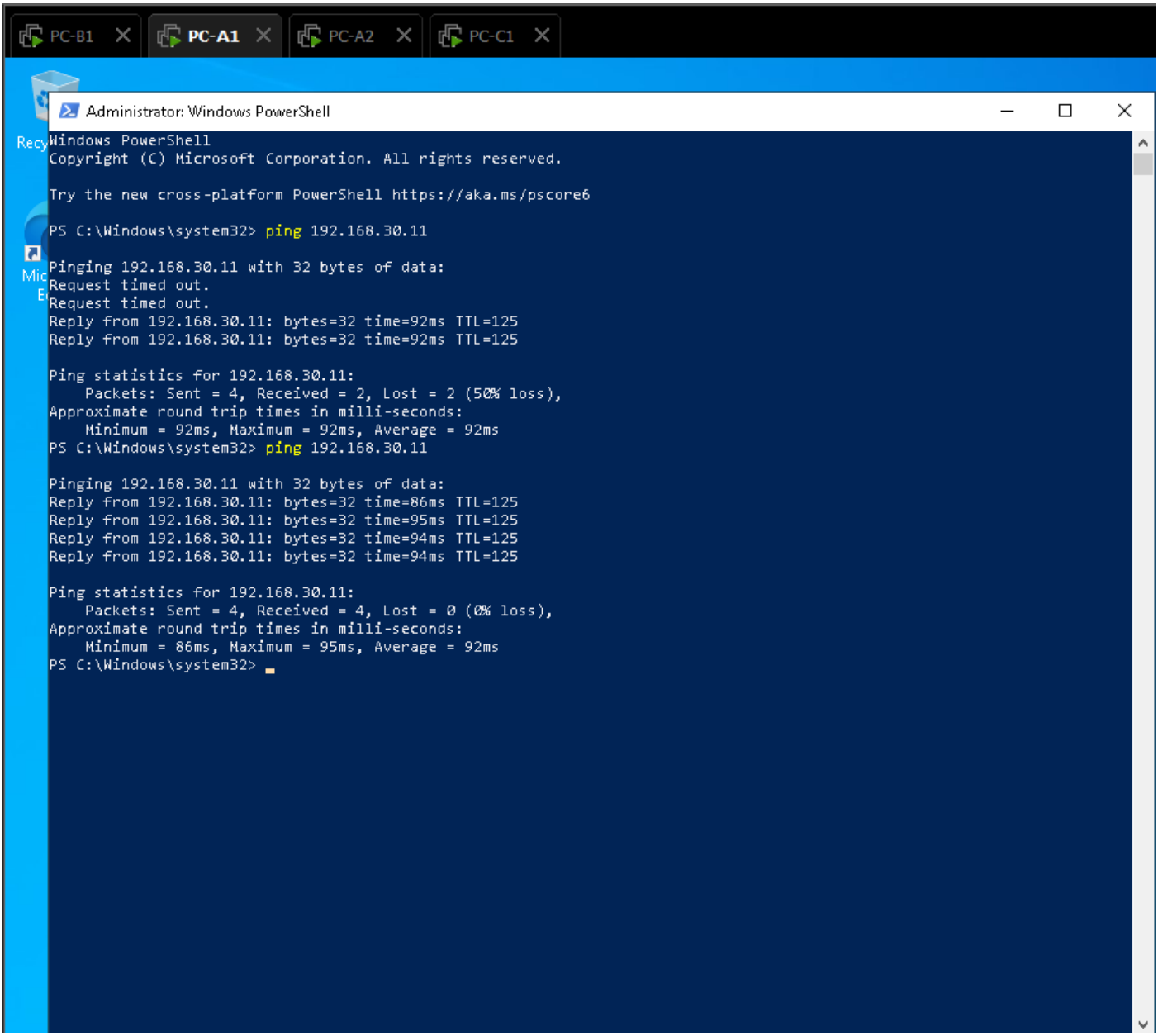
Captura Wireshark

filtro icmp en el enlace R1–R2, Echo Request/Reply



Captura Wireshark – Parte 2

Resultado en la máquina virtual.



Falla provocada

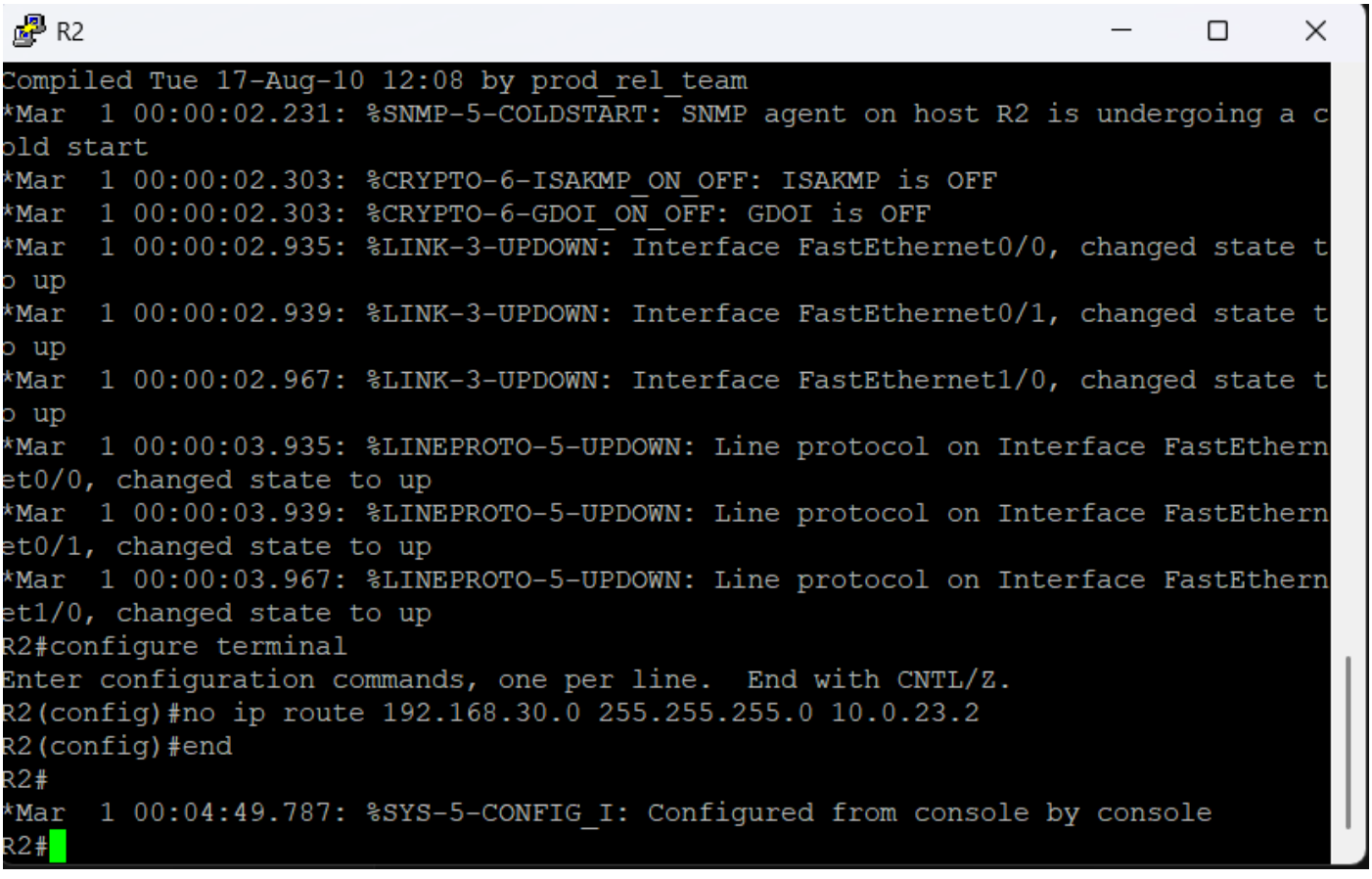
sin la ruta a LAN-C el ping muere; restaurada, responde

```
PS C:\Windows\system32> hostname
PC-A1
PS C:\Windows\system32> ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:
Reply from 10.0.12.2: Destination host unreachable.
Reply from 10.0.12.2: Destination host unreachable.
Reply from 10.0.12.2: Destination host unreachable.
Reply from 10.0.12.2: Destination host unreachable.

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
PS C:\Windows\system32>
```

Consola R2 – GNS3



Recuperación

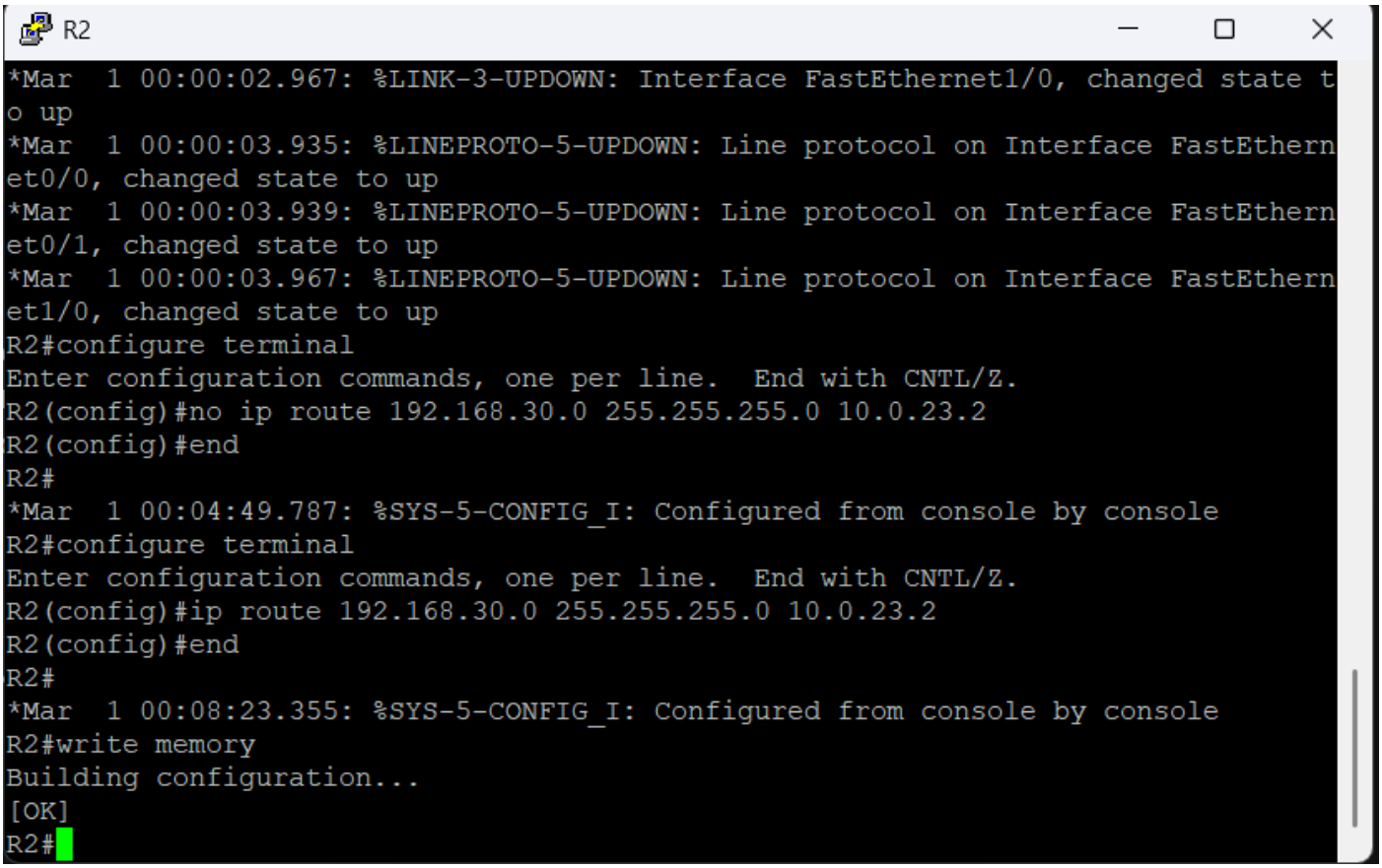
Prueba restaurada en VM Ware

```
PS C:\Windows\system32> hostname
PC-A1
PS C:\Windows\system32> ping 192.168.30.11

Pinging 192.168.30.11 with 32 bytes of data:
Reply from 192.168.30.11: bytes=32 time=85ms TTL=125
Reply from 192.168.30.11: bytes=32 time=94ms TTL=125
Reply from 192.168.30.11: bytes=32 time=94ms TTL=125
Reply from 192.168.30.11: bytes=32 time=93ms TTL=125

Ping statistics for 192.168.30.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 85ms, Maximum = 94ms, Average = 91ms
PS C:\Windows\system32>
```

Prueba restaurada en R2 Consola – GNS3



CONCLUSIÓN

La práctica deja claro cómo se comporta el enrutamiento estático. Cada router solo conoce sus redes directas, así que hay que decirle a mano el siguiente salto para cada red remota, tanto de ida como de vuelta. La **ACL** extendida trabajó de forma quirúrgica: cortó el **ICMP** de **LAN-B** hacia **LAN-C** sin tocar el resto del tráfico. La falla provocada en **R2** mostró lo que pasa cuando una ruta manual desaparece: se borró una sola línea y la conectividad extremo a extremo se cayó, con el propio router reportando el destino como inalcanzable. Restaurada la ruta, todo volvió a responder. Con esto queda montada la base de configuración de routers **Cisco**, direccionamiento **IPv4** y diagnóstico de fallas de conectividad.

MUCHAS GRACIAS POR LEER – (LUISMARTINPM)

